PROJECT REPORT: ANALOG FUNCTION GENERATOR

Hardware Design, Schematic Analysis & Oscilloscope Verification Report
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1. EXECUTIVE SUMMARY

This report documents the end-to-end design, simulation, breadboard prototyping, and custom PCB fabrication of a precision Multifunction Analog Function Generator supporting Sine, Triangle, and Square waveforms from 10 Hz to 100 kHz with adjustable duty cycle and variable amplitude output.

2. HARDWARE ARCHITECTURE

* Core Generator: ICL8038 / XR2206 monolithic waveform generator IC with dual-rail op-amp buffers
* Power Regulation: Dual +/- 12V low-noise linear regulation (LM7812 / LM7912) with transient filtration
* Output Stage: High slew-rate operational amplifiers with 50-ohm output impedance match

3. VERIFICATION & MEASUREMENTS

* Total Harmonic Distortion (THD) on sine output: < 1.2% measured with digital storage oscilloscope
* Square wave rise/fall time: < 22 nanoseconds into 50-ohm test load
* Amplitude range: 0.1 Vpp to 10.0 Vpp continuously adjustable

4. CONCLUSION

The hardware meets all target electrical specifications and operates stably under continuous load.